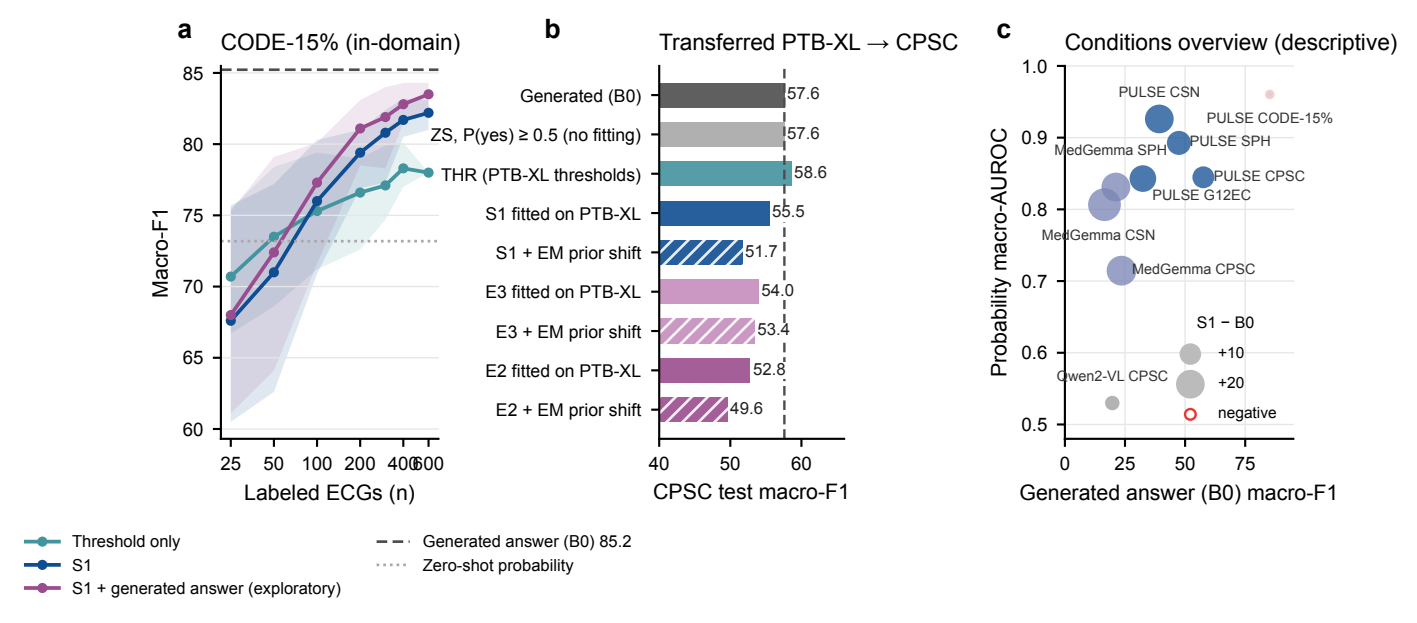
a
CODE-15% (in-domain)
Macro-F1
85
80
75
70
65
60
25
50
100
200
400
600
Labeled ECGs (n)
b
Transferred PTB-XL → CPSC
Generated (B0)
ZS, P(yes) ≥ 0.5 (no fitting)
THR (PTB-XL thresholds)
S1 fitted on PTB-XL
S1 + EM prior shift
E3 fitted on PTB-XL
E3 + EM prior shift
E2 fitted on PTB-XL
E2 + EM prior shift
57.6
57.6
58.6
55.5
51.7
54.0
53.4
52.8
49.6
40
50
60
CPSC test macro-F1
c
Conditions overview (descriptive)
Probability macro-AUROC
1.0
0.9
0.8
0.7
0.6
0.5
PULSE CSN
PULSE CODE-15%
PULSE SPH
MedGemma SPH
PULSE CPSC
PULSE G12EC
MedGemma CSN
MedGemma CPSC
Qwen2-VL CPSC
S1 − B0
+10
+20
negative
0
25
50
75
Generated answer (B0) macro-F1
Threshold only
S1
S1 + generated answer (exploratory)
Generated answer (B0) 85.2
Zero-shot probability